IS264 Principles of Database Systems

Semester I, 2024/2025

Tutorial Questions

Topic: Entity Relationship Diagram

Question One:

The owners of wineries at Cetawico Hombolo in Dodoma want to keep track of the different wines, wine-growing estates, and wine dealers. For this, they set up a database containing the following relations:

In the table WINE the following data about wines are stored: the official control number of the wine (OCNo), the name of the wine (WName), and the name of the wine-growing estate where the wine is coming from (WEstate). Furthermore, the year of the wine, its alcoholic content (Percentage, e.g., 5.2%), and its quality are stored (e.g. 1 -excellent, 2 – expensive, etc).

The table ESTATE contains information about the wine-growing estate, its region, the year of the foundation, and the average wine quality. The table WDEALER stores the names and addresses of wine dealers.

Finally, the table OFFERS stores information about wines offered by wine dealers and the price each dealer charges for that wine.

(i) List down all the entities.

(ii) Identify and list down all the primary keys for each entity.

(iii) Draw an ER diagram.

Question Two

A university consists of several departments. Each department (characterized by department code, name, and location) offers several courses. Several modules make up each course. Each course is characterized by a course code and name. Students enroll in a particular course and take modules toward the completion of that course. Each student is characterized by student id (Sid) and student name. Each module (characterized by module code, module name, and credit units) is taught by a lecturer from the appropriate department, and each lecturer (characterized by ssn, staff name, address, salary, etc) tutors a group of students.

(i) Draw an entity relationship diagram corresponding to the given scenario.

Questions Three

James Bond wants to set up a database for its members and trainers. One of the aims is to record information about which members participated in which sessions and which trainers supervised which sessions.

Mr Bond operates various fitness centres in various cities. Every centre is characterized by a unique name (*Fname*) (e.g., Fitplaza, my6pack) and *location*. Every centre has an *address* and one or more rooms (you can consider the address as atomic). Every room has a maximum capacity. Within a centre, each room has a unique number (*rID*) such as 1, 2, 3, etc. People can register for individual or group sessions in different centres. Each group session requires exactly one trainer. Individual sessions are done without a trainer. For each person, we want to store the *first name*, *family name*, and *birth date*. You can assume that the combination of *first name, family name*, and *birth date* is unique. For each trainer, the diploma is also recorded.

A person can be a trainer in one session and a participant in another session (either individual or group session). The model should also include information about people (e.g., prospects) that have not participated in any sessions yet, or trainers (e.g., interns) that have not supervised any group sessions yet.

For each session, the *date* and *starting hour* should be recorded. For group sessions, also the *type* should be stored (e.g., aerobics, body styling, etc.). Sessions can start at the same time on the same day but in different rooms of a centre or different centres. At a given start hour of a given day, at most one individual or group session can start in a given room of a given centre.

(iv) List down all the entities.

(v) Identify and list down all the primary keys for each entity.

(vi) Draw an ER diagram.